

Mathematical modeling of a modular convection-enhanced evaporation system

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HIGHLIGHTS

- A modular brine management method for decentralized plants is proposed.
- Mathematical model of a convection-enhanced evaporation system is presented.
- Active control of liquid temperature and air speed improve the evaporation performance up to 13 times.
- The negative effect of high relative humidity can be mitigated by liquid preheating.

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ABSTRACT

This study presents mathematical modeling of a novel convection-enhanced evaporation (CEE) system. The proposed system represents a modular brine management option for decentralized desalination plants and small-scale industries. Liquid is pumped over horizontally packed evaporation surfaces and forced convection is induced by means of a fan. The developed model of heat and mass transfer in the CEE system is solved using the finite difference method to predict the evaporation rate and the spatial distribution of humidity and temperature, based on operating condition (ambient temperature and relative humidity, air flow speed, and the spacing between the evaporation surfaces) and liquid inlet condition (temperature, flow rate and salinity). Comparison between model-predicted evaporation rates and experimental results show good agreement; for 11 out of 16 operating conditions error was within 10%, for 4 conditions error was within 20%, and for 1 condition error was within 35%. The effect of various operating conditions on the evaporation performance is also investigated. An increase in evaporation of up to 13 times is achieved by active control of liquid inlet temperature and air flow speed. Preheating the liquid was found to be a good strategy to overcome the negative effect of high relative humidity.

1. Introduction

Desalination systems provide alternative water sources, helping to address the global scarcity of freshwater. However, all desalination processes result in two fluid streams, the product stream and the byproduct stream known as the brine or concentrate [1]. Heavily contaminated brines are also produced in several other industries, such as textile dyeing and electroplating. Poor management of brine effluent can lead to negative environmental impacts due to the high concentration of dissolved salts and other chemical contaminants [2–4]. Estimates show that brine disposal cost ranges from 5 to 33% of the total desalination cost depending on the contaminants present [5,6]. Regulations

increasingly mandate industries to treat brine effluents to reach zero liquid discharge [7], further increasing the cost and energy demand of brine management. Additionally, most reviews of inland brine management techniques focus on large facilities processing greater than 100 m³/day brine. For small-volume brine producers, like those targeted in this study, the expected specific cost and energy for treatment will be higher than that cited in the literature, as small size significantly works against thermal efficiency. There is an urgent need for environmentally sustainable, energy efficient, and cost-effective brine management methods especially for small-scale, decentralized desalination plants and industrial systems.

Brine management methods can be categorized in two ways, (1) as

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brine disposal methods in which brine is discarded as waste, and (2) as brine treatment methods in which the aim is to minimize brine volume, recover freshwater, and/or produce salt byproducts. Brine is commonly disposed of via several methods including surface water discharge, sewer water discharge, deep well injection, land application, and evaporation ponds (Fig. 1). These brine disposal methods are unsustainable and may have negative environmental impacts such as pollution of marine environment, groundwater pollution, and soil salinization, in addition to being cost prohibitive [1]. Brine treatment methods, such as multi-effect distillation, generally mitigate issues of direct environmental pollution but are associated with high energy consumption and upfront cost, which limits their implementation and adoption.

Despite potential environmental impacts, conventional evaporation ponds remain an attractive alternative as they exploit the natural drying power of wind to evaporate water, leaving solid contaminants behind. Evaporation from ponds occurs due to the positive difference in vapor pressure between the liquid surface and the flowing air (Eq. (1), Fig. 2b). This relationship is demonstrated in the general equation governing evaporation from a body of water [8]:

$$E_{vap} = N_e \bar{u}_r (\bar{e}_w^* - \bar{e}_a), \text{ where,} \quad (1)$$

$$N_e = 3.367 \times 10^{-9} A^{-0.05}, \quad (2)$$

and E_{vap} is the evaporation rate, $\bar{u}_r$ is the time-averaged wind speed, $\bar{e}_w^*$ and $\bar{e}_a$ are the vapor pressure of water at its surface temperature (saturation pressure) and the partial pressure of water vapor in air flowing above the surface, respectively, N_e is the wind coefficient, and A is the area of the evaporation body. Mass transfer (Eq.(1)) drops off significantly with the increasing extension of the water body downwind (Eq. (2)); this is because the vapor pressure in the air is increased due to the evaporation that occurs in the previous segment of the pond.

Wind-aided intensified evaporation (WAIV) was proposed by Gilron et al. to mitigate the large land area requirement of evaporation ponds [9]. WAIV is a brine volume reduction method that recirculates the brine as a falling film over vertically packed fabric surfaces while wind blows across the surfaces to evaporate water. This configuration increases the evaporation capacity per unit area of footprint by a factor of 10 compared to a standard evaporation pond [9]. Arnal et al. also investigated increased surface area as a mean of enhancing natural evaporation with the use of capillary absorbents [5].

In this text, the approach of evaporating brine from packed wetted surfaces by convection is referred to generally as *convection-enhanced evaporation* (CEE). In current implementations of CEE (e.g. [5,9]) wind availability limits evaporation rate. Additionally, critical parameters affecting evaporation performance such as the air flow speed, spacing between the surfaces, and brine flow rate need further critical investigation to be understood and optimized; the current literature is limited to experimental testing [5,9,10,11] with no clear mathematical model that combines the effect of all critical parameters.

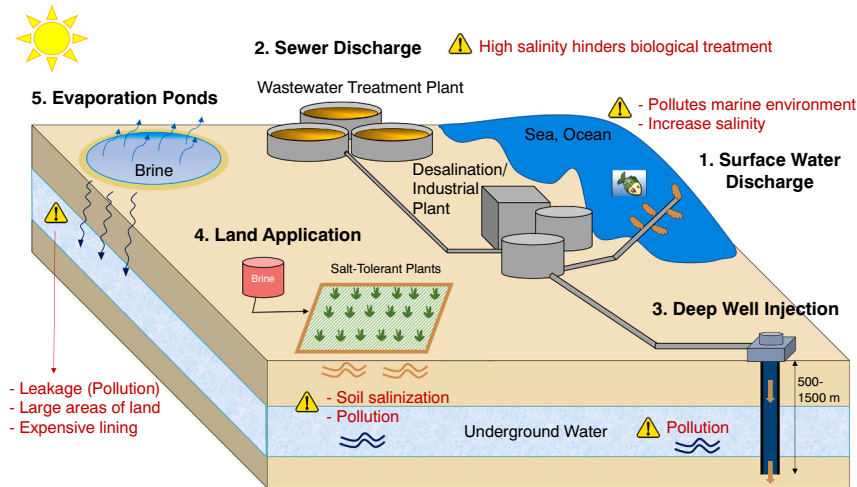


Fig. 1. Common brine disposal methods and the associated environmental challenges.

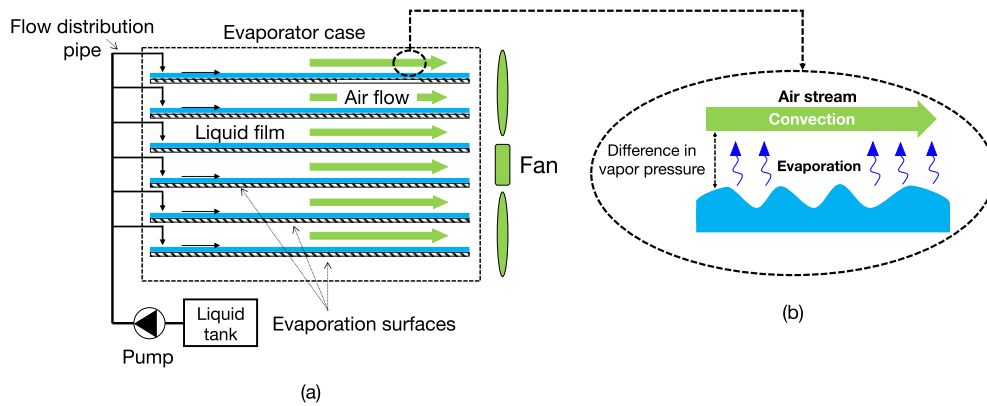


Fig. 2. (a) Schematic for the proposed convection-enhanced evaporation (CEE) system. A flow distribution pipe releases liquid over each evaporation surface forming a thin film. A fan induces air flow through the evaporator. The evaporation capacity of CEE can be increased by packing more evaporation surfaces in parallel. (b) Liquid evaporation by air convection occurs due to the difference in vapor pressure between the liquid surface and the flowing air.

A schematic diagram showing an alternative implementation of a CEE system is shown in Fig. 2. The evaporation system is comprised of a set of horizontally packed hydrophilic evaporation surfaces at uniform vertical spacing. The capacity of the evaporation system can be increased by adding additional evaporation surfaces, making the system modular and easy to fabricate. Liquid is uniformly released by means of a flow distribution pipe and orifices along the width of each evaporation surface forming a thin film. A fan is fixed at the opposite side of the liquid injection as a driving force for air flow. As air flows over the liquid film, the difference in vapor pressure between the air and liquid surfaces induces evaporation (Fig. 2b). The humid air leaves the evaporator, and any remaining liquid that is not evaporated is recycled to the tank.

The operating principle behind CEE systems is similar to that of a cooling tower where hot water is sprayed at the top of the tower and flowing air cools the water by evaporation. The heat and mass transfer in a cooling tower is commonly modeled according to one of three methods: the Merkel method [12], the Poppe method [13] and the effectiveness-NTU (e-NTU) method [14]. The Merkel theory, developed in 1925, is not expected to adequately predict the evaporative performance in the CEE systems of interest because it, (1) neglects the reduction in water flow rate due to evaporation in the energy balance equation, and (2) assumes a Lewis factor equal to unity in order to reduce the solution to a simple hand calculation [12,15]. The e-NTU method utilizes an analogy between the analysis of heat exchangers and the evaporative process [14]. This simplifies the numerical procedure, but it is still based on the same two assumptions as in the Merkel method. The Poppe method was developed in the 1970s and solves the simultaneous heat and mass transfer equations numerically without making the simplifying assumptions of the Merkel method [13]. The equations are based on mass and energy balances accompanied with interphase heat and mass transfer. This method accounts for the decrease in water flow rate due to evaporation and the water content of the exiting air. Kloppers and Kröger presented a detailed model for counter-flow and cross-flow wet cooling towers based on the Poppe method and presented the results in comparison to the Merkel and e-NTU methods [16,17]. The Poppe method resulted in more accurate results when compared to experimental data. They concluded that the Poppe method is more complex than Merkel and e-NTU methods but more accurate especially if the amount of water evaporated and exiting air temperature are important in the design.

Other studies of combined heat and mass transfer modeling for liquid films are presented in several desalination systems. Soufari et al. studied the evaporation from humidification-dehumidification desalination where the humidifier and the dehumidifier are made up of structural packing [18]. The presented model was utilized to optimize the system for minimum specific thermal energy consumption, maximum ratio of distilled water to feed water flow rate, and maximum heat recovery. Zamen et al. presented a similar model for a solar humidification-dehumidification desalination unit for freshwater production from seawater in a greenhouse [19]. To increase the mass transfer driving force, they increased the hot water temperature in the humidifier by means of a solar collector. Najim et al. presented a study for the evaporation of a saline liquid film from a heated vertical tube [20]. The simulation allowed for the investigation of the effect of liquid salinity on evaporation performance.

This work presents mathematical modeling of a CEE system. Evaporation by convection from horizontally packed flat surfaces is studied. A formulation similar to the Poppe method is utilized to develop a generalized simulation model, solved numerically using the finite difference method. The presented model is capable of predicting the evaporation rate and the spatial distribution of humidity and temperature along the CEE system based on the ambient conditions (temperature and relative humidity), the liquid inlet conditions (temperature, flow rate and salinity), and the air flow speed. The model is utilized to investigate the effect of these conditions on evaporation performance; model-predicted performance is compared to experimental performance

from a pilot CEE system operating in outdoor conditions. The presented system configuration represents a modular evaporation system, which makes it a suitable solution for brine management for decentralized desalination plants and small industries.

2. Mathematical model

The proposed mathematical model is developed to analyze the mass and heat transfer in the CEE system (Fig. 2) with horizontal wetted evaporation surfaces (Fig. 3). The governing equations are formulated under the following assumptions:

1. Steady state, one dimensional heat and mass transfer.
2. Heat and mass are transferred in the direction normal to the evaporation surface. Diffusion of heat and vapor in the x-direction is neglected.
3. Fully developed flow is considered. Viscous dissipation and spatial variations in pressure are neglected.
4. The evaporation surface is entirely wetted by the liquid film, and the film is evenly distributed along the width (W) of the surface.
5. Liquid film is thin, so the thermal resistance on the liquid side is negligible. As a result, the local temperature of the liquid-gas interface is assumed equal to the local temperature of the liquid.
6. Air at the interface is saturated at the local liquid temperature.
7. Air is treated as an ideal gas.
8. All phases are in thermodynamic equilibrium and homogeneous flow is assumed.
9. Uniform heat and mass transfer coefficients assumed in each control volume.

2.1. Governing equations

The general governing equations for mass, momentum and energy conservation for a Newtonian fluid are:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (3)$$

$$\frac{\partial (\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) = \rho \mathbf{f} + \nabla \cdot \mathbf{T} \quad (4)$$

$$\frac{\partial \rho \left(e + \frac{\mathbf{u} \cdot \mathbf{u}}{2} \right)}{\partial t} + \nabla \cdot \left(\rho \mathbf{u} \left(e + \frac{\mathbf{u} \cdot \mathbf{u}}{2} \right) \right) = -\nabla \cdot \mathbf{q}'' + \rho \mathbf{f} \cdot \mathbf{u} + \nabla \cdot (\boldsymbol{\tau} \cdot \mathbf{u}) + \dot{q} \quad (5)$$

The conservation of momentum equation, Eq. (4), is dotted by the velocity vector and subtracted from the conservation of energy equation, Eq. (5). The conservation of energy equation becomes

$$\frac{\partial (\rho i)}{\partial t} + \nabla \cdot (\rho \mathbf{u} i) = -\nabla \cdot \mathbf{q}'' + \nabla \mathbf{u} : \boldsymbol{\tau} + \frac{\partial p}{\partial t} + \mathbf{u} \cdot \nabla p + \dot{q}. \quad (6)$$

After applying the listed assumptions, the governing equations are simplified to:

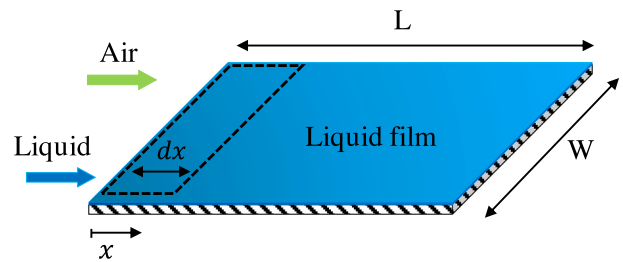


Fig. 3. An evaporation surface showing a differential element. Liquid brine is released over the surface forming a thin liquid film. Air velocity is in the same direction as liquid velocity.

$$\frac{d(\rho \mathbf{u})}{dx} = 0 \quad (7)$$

$$\frac{d(\rho \mathbf{u} i)}{dx} = -\frac{d\mathbf{q}''}{dx} \quad (8)$$

Notice that subtracting the conservation of momentum equation from the conservation of energy equation and applying assumption (3) omit the conservation of momentum equation from the final set of governing equations. As the mass flow rates of air and liquid are the measured variables, Eqs. (7) and (8) are converted to the mass flow rate form by multiplying with the cross-sectional area of the flow, such that

$$\frac{d\dot{m}}{dx} = 0, \text{ and,} \quad (9)$$

$$\frac{d(\dot{m} i)}{dx} = -\frac{d\mathbf{q}}{dx} \quad (10)$$

The conservation of mass, Eq. (9), for the control volume in Fig. 4 is expressed as

$$d\dot{m}_l = -\dot{m}_a d\omega_g, \quad (11)$$

where $\dot{m}_l$ and $\dot{m}_a$ are the liquid and dry air mass flow rates respectively, and ω_g is the moisture content of the moist air. This equation states that the rate of water evaporation from the liquid is equal to the rate of moisture added to the air. The driving force for mass transfer is the difference in moisture content between the air layer immediately adjacent to the interface and the mean stream air. The rate of mass transfer at the interface is given by

$$d\dot{m}_l = -k(\omega_{int} - \omega_g) dA, \quad (12)$$

where k is the mass transfer coefficient, and ω_{int} is the moisture content of the air layer immediately adjacent to the interface. Substituting Eq. (12) into Eq. (11) yields

$$d\omega_g = \frac{k}{\dot{m}_a} (\omega_{int} - \omega_g) dA. \quad (13)$$

The conservation of energy, Eq. (10), for the air side of the control volume in Fig. 4 is expressed as

$$\dot{m}_a di_g = h(T_l - T_g) dA + i_v k(\omega_{int} - \omega_g) dA, \quad (14)$$

where i_g is the enthalpy of the moist air, h is the heat transfer coefficient, T_l and T_g are the temperatures of the liquid and the moist air respectively, and i_v is the enthalpy of water vapor. Eq. (14) states that the rate of change of air enthalpy is due to the combination of the simultaneous sensible (convection) and latent (evaporation) heat transfer occurring at the air-liquid interface. The enthalpy of moist air is related to its local temperature by

$$i_g = c_{p,a}(T_g - 273.15) + \omega_g [i_{fgw0} + c_{p,v}(T_g - 273.15)], \quad (15)$$

where $c_{p,a}$ and $c_{p,v}$ are the specific heats of dry air and water vapor

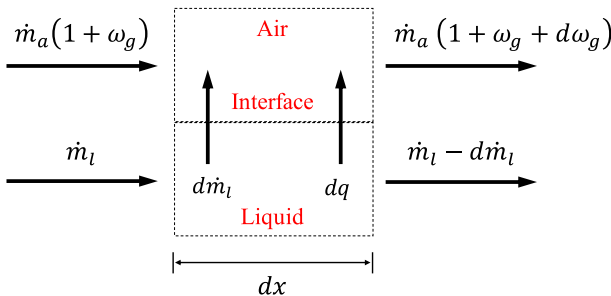


Fig. 4. Control volume from the evaporation surface showing the mass and heat transfer.

respectively, and i_{fgw0} is the latent heat of water evaluated at 0 °C. Specific heats in Eq. (15) are evaluated at $(T_g(K) - 273.15)/2$.

The conservation of energy, Eq. (10), for the entire control volume in Fig. 4 is expressed as

$$c_{p,l} \dot{m}_l dT_l + c_{p,l} T_l d\dot{m}_l + \dot{m}_a di_g = 0, \quad (16)$$

where $c_{p,l}$ is the specific heat of the liquid. Substituting Eq. (11) into Eq. (16) yields

$$dT_l = \frac{\dot{m}_a}{\dot{m}_l} \left(T_l d\omega_g - \frac{1}{c_{p,l}} di_g \right) \quad (17)$$

Substituting Eq. (14) into Eq. (17) yields

$$dT_l = \frac{dA}{\dot{m}_l c_{p,l}} [k(\omega_{int} - \omega_g)(c_{p,l} T_l - i_v) - h(T_l - T_g)] \quad (18)$$

Finally, the accumulated evaporation rate at location x along the surface is the difference in liquid flow rate expressed as

$$E_{vap,x} = \dot{m}_{l,in} - \dot{m}_{l,x}, \quad (19)$$

where $E_{vap,x}$ is the accumulated evaporation rate at location x , and $\dot{m}_{l,in}$ and $\dot{m}_{l,x}$ are the liquid mass flow rates at the inlet and at location x respectively. Air mass flow rate is related to air flow speed as

$$\dot{m}_a = \rho u W d, \quad (20)$$

where ρ is the air density, u is the air speed, W is the evaporation surface width, and d is the vertical spacing between evaporation surfaces. The local salinity of the liquid at position x along the surface is expressed as

$$S_x = S_{in} \frac{\dot{m}_{l,in}}{\dot{m}_{l,x}}, \quad (21)$$

where S_{in} and S_x are the liquid salinity at the inlet and at location x respectively.

Eqs. (12), (13), (14) and (18) form the governing ordinary differential equations (ODE) for the CEE system.

2.2. Boundary conditions and closure models

The boundary conditions for the final governing equations are:

(1) Inlet air temperature and moisture content,

$$x = 0 : T_g = T_{g,in}, \omega_g = \omega_{g,in}$$

(2) Inlet liquid temperature, flow rate and salinity,

$$x = 0 : T_l = T_{l,in}, \dot{m}_l = \dot{m}_{l,in}, S = S_{in}$$

The mass transfer coefficient (k) is calculated from the Lewis factor relation [21]

$$k = \frac{h}{c_{p,g} Le_f}, \quad (22)$$

where the specific heat of the moist air ($c_{p,g}$) is assumed a linear function of moisture content given as

$$c_{p,g} = c_{p,a} + \omega_g c_{p,v}, \quad (23)$$

and the Lewis factor (Le_f) is given according to Bosnjakovic [21,22,23] as

$$Le_f = 0.865^{2/3} \frac{\left(\frac{\omega_{int} + 0.622}{\omega_g + 0.622} - 1 \right)}{\ln \left(\frac{\omega_{int} + 0.622}{\omega_g + 0.622} \right)}. \quad (24)$$

The heat transfer coefficient (h) is calculated from Dittus-Boelter

equation [24],

$$h = 0.023 Re^{0.8} Pr^n \left(\frac{\lambda}{D_h} \right), \quad (25)$$

where Re is the Reynolds number, Pr is the Prandtl number of air, λ is the thermal conductivity of air, D_h is the hydraulic diameter, and the exponent n is equal to 0.4 when the liquid local temperature is higher than gas local temperature, and is equal to 0.3 otherwise. The Reynolds number is given as

$$Re = \frac{u D_h}{\nu}, \quad (26)$$

where ν is the kinematic viscosity of air, and the hydraulic diameter (D_h) is given as

$$D_h = \frac{2 W d}{W + d}. \quad (27)$$

The enthalpy of water vapor, i_v , is for calorically ideal gases a linear function of its local temperature, given by

$$i_v = i_{g,w0} + c_{p,v}(T_l - 273.15), \quad (28)$$

where the specific heat of water vapor in Eq. (28) is evaluated at $(T_l(K) - 273.15)/2$.

The moisture content of saturated air flowing over the liquid film interface can be calculated from the water vapor saturation pressure as

$$\omega_{int} = 0.622 \frac{P_{sat}}{p_{atm} - P_{sat}}, \quad (29)$$

where p_{sat} and p_{atm} are the water vapor saturation pressure and the atmospheric pressure respectively. Water vapor saturation pressure for temperature range of 273 K (0 °C) to 473 K (200 °C) is calculated as [25]

$$\ln(p_{sat}) = \frac{a_1}{T_{int}} + a_2 + a_3 T_{int} + a_4 T_{int}^2 + a_5 T_{int}^3 + a_6 \ln(T_{int}), \quad (30)$$

where,

$$\begin{aligned} a_1 &= -5800.2206, & a_2 &= 1.3914993, \\ a_3 &= -0.048640239, & a_4 &= 0.41764768 \times 10^{-4}, \\ a_5 &= -0.14452093 \times 10^{-7}, & a_6 &= 6.5459673, \\ p_{atm} &= 101325 Pa \end{aligned}$$

with the interface temperature (T_{int}) equal to the local liquid temperature (K).

The presented formulation is for pure water. To account for salinity, the thermophysical properties of pure water are replaced with those for saline liquid. Sharqawy et al. [26] provide a review of correlations for each property along with their associated ranges of validity and accuracy; Refer to Appendix C for the correlations used in this study.

2.3. Numerical solution

The finite difference method is utilized to solve the governing differential equations. The domain is discretized into N_x grid points in the x-direction, and the fourth order Runge-Kutta method is utilized to solve the governing equations, Eqs. (12), (13), (14) and (18), for the spatial distribution of the enthalpy of moist air (i_g), moisture content (ω_g), liquid temperature (T_l), and the remaining liquid flow rate after evaporation ($\dot{m}_l$). The moist air temperature (T_g) is updated at each grid point according to Eq. (15). Refer to the Appendix A for the numerical procedure of the Runge-Kutta method.

2.4. Power consumption

The frictional pressure drop is given by the Darcy-Weisbach equation as

$$\Delta p = f \frac{L}{D_h} \frac{\rho u^2}{2}, \quad (31)$$

where L is the evaporation surface length. The Darcy friction coefficient (f) for a rectangular flow geometry is given by an analytical correlation as [27],

$$f = \frac{96}{Re} \times \left(1 - 1.3553\alpha + 1.9467\alpha^2 - 1.7012\alpha^3 + 0.9564\alpha^4 - 0.2537\alpha^5 \right), \quad (32)$$

where α is a dimensionless shape factor given as

$$\alpha = \frac{d}{W}. \quad (33)$$

The phase change occurring during evaporation creates higher resistance to flow resulting in higher pressure drop. A multiphase-correction factor of 3 is commonly utilized [27,28] and was found to match the current experimental results within 4%.

The electric power consumption per evaporation surface due to fan operation is estimated as

$$P_{el} = \frac{\Delta p \dot{V}_a}{\eta_{fan}}, \quad (34)$$

where $\dot{V}_a$ is the air volumetric flow rate given by

$$\dot{V}_a = u W d, \quad (35)$$

and η_{fan} is the fan efficiency. In this study, a fan efficiency of 67% is assumed [28].

The thermal power consumption per evaporation surface due to liquid preheating is estimated as

$$P_{th} = \dot{m}_l c_{p,l} \Delta T, \quad (36)$$

where ΔT is the temperature rise due to heating.

3. Experimental validation

In order to evaluate the model predictions, modeled evaporation rate was compared to data provided by QuadSun Solar Solutions for a pilot CEE system. The objective was to assess the accuracy of the model under real operating conditions, without deriving new empirical parameters. The experiments were performed at the company headquarters in outdoor conditions in Gurgaon, India. Data provided by QuadSun included operating conditions (ambient temperature, relative humidity, liquid inlet temperature, liquid flow rate, and air speed), and performance metrics (evaporation rate, exit air temperature, and exit relative humidity). The provided experimental operating conditions were implemented as input boundary conditions in the model simulations.

Fig. 5 shows the experimental setup. Horizontal evaporation surfaces were stacked with a vertical separation distance of 2.4 cm. Each surface had dimensions of 0.6 m $\times$ 0.6 m. A liquid distribution system made up of 3.4 mm (inner diameter) metal pipeline with 23 evenly spaced 1 mm orifices was arranged at the top of the leading edge of each evaporation surface to release liquid along the width of the surface. A pump was used to circulate liquid between a recirculation tank and the surfaces. To control the liquid flow rate, liquid was injected onto the evaporation surfaces by cycling the pump on and off for a prescribed amount of time. For example, to achieve a liquid flow rate of 6.5 L/h/m²_{surface}, the pump was operated in a cycle of 3 s of run time every 35 s. All experiments were run with local tap water having a total dissolved salinity of less than 500 mg/L. An industrial axial fan fixed at the front of the setup (in pull configuration) was used to induce air flow along the evaporator at constant speed (6 m/s to 10 m/s, fan motor rated power of 300 W). Liquid that was not evaporated was drained back to the recirculation

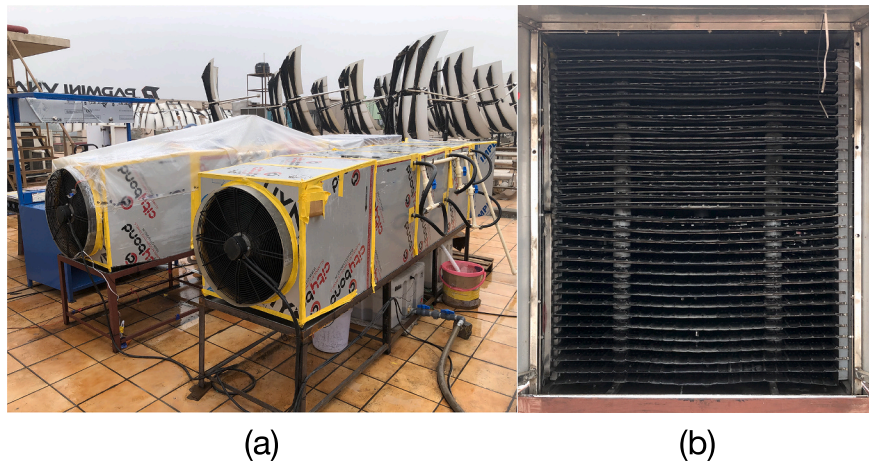


Fig. 5. (a) Photograph of an experimental CEE system constructed by QuadSun Solar, Ltd. Liquid is pumped over horizontally packed surfaces and an axial fan is utilized to induce air flow through the evaporation surfaces. (b) The horizontal arrangement of the evaporation surfaces.

tank. Experimental duration ranged from 30 min to 180 min over a wide range of ambient conditions and liquid injection flow rates and temperatures: ambient temperature (8 °C to 30 °C), relative humidity (40% to 80%), liquid temperature (13 °C to 66 °C), and liquid mass flow rate (0.8 L/h/m²_{surface} to 6.5 L/h/m²_{surface}). The evaporation rate from the experiment was normalized by the number of evaporation surfaces for the purpose of comparison with the mathematical model. Refer to [Appendix B](#) for a summary of experimental conditions and results.

The experimental evaporation rates were calculated as

$$E_{vap} = \frac{\Delta V}{\Delta t N_s}, \quad (37)$$

where ΔV is the change in volume in the recirculation tank, Δt is the experiment duration and N_s is the total number of evaporation surfaces. The reported air speed was obtained by averaging measurements done near the corners and center of the packed surfaces. The temperature and relative humidity of the air were recorded at the inlet and exit sides of the CEE for each experiment. Accuracy, range, and resolution of the measurement instruments are presented in [Table 1](#).

[Fig. 6](#) shows a comparison between the experimental and model-predicted values for evaporation rate per surface, exit air relative humidity, and exit air temperature. Good agreement was obtained between the model and the experimental results; for 11 out of 16 operating conditions relative error was within 10%, for 4 conditions relative error was within 20%, and for 1 condition, relative error was within 35%. The relative humidity and temperature of the exit air agreed with the experiments to within 3% and 2 °C, respectively ([Table B.2](#)).

Deviation between the model and the experimental results was expected due to differences between the assumptions made in [Section 2](#) and the real world operating scenario. [Assumption \(4\)](#) requires the evaporation surface to be covered with a liquid film for the entire operation time. However, uneven flow from the orifices due to clogging and/or pressure variations along the length of the flow distribution pipe

may have caused variation in the liquid flow rate and resulted in uneven wetting of the surfaces. Moreover, the cyclic operation of the liquid injection system, as described above, also resulted in unsteady, partial wetting of the surfaces. In addition, several surface materials of different hydrophilic properties were utilized. The hydrophilic nature of the surface may affect its wettability and consequently the formed film, with nylon visually providing a more even distribution of liquid than the polypropylene surfaces. Although these conditions deviate from the developed model, the model still provides very good predictive capabilities when compared to the experimental results. Furthermore, the uncertainty of the measuring devices utilized during the experiment, and the dynamic variation of the operating parameters due to outdoor testing likely contributed to differences between model predictions and experimental readings.

4. Results and discussion

In this section the developed mathematical model ([Section 2](#)) is used to examine the parametric relationship between operating conditions and evaporation performance. We begin by defining a *base case* condition, for which the system is operated under the conditions listed in [Table 2](#). These conditions are then varied in the following discussion to investigate their effect on evaporation performance. The difference between liquid and ambient inlet temperatures is referred to as ΔT ($\Delta T = T_{l, in} - T_{g, in}$). Throughout this investigation, the liquid inlet temperature will be varied while keeping the ambient temperature constant as listed in [Table 2](#). The reported evaporation rate in all figures is the evaporation rate for a single evaporation surface having an area of one square meter. This value should be multiplied by the number of horizontally packed evaporation surfaces in a single module to calculate the evaporation capacity per unit footprint.

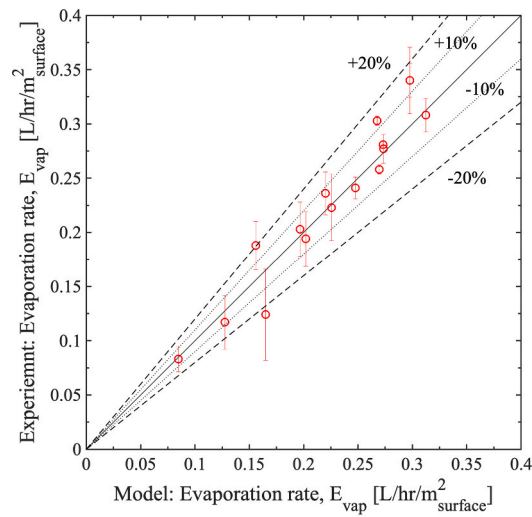
A grid independence study was conducted for the developed model to identify the optimal grid resolution. [Fig. 7](#) shows the result of this study. The total evaporation rate per surface was computed at different grid resolutions ranging from 10 nodes to 600 nodes. It was identified that grid independent results were obtained at 500 nodes or higher. Therefore, for the subsequent analysis presented in this work, a computational grid of 500 nodes is utilized.

4.1. Spatial profiles

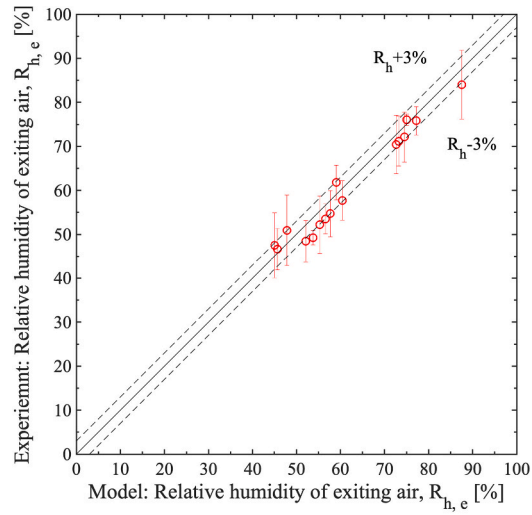
Starting with the base case conditions ([Table 2](#)), the spatial distribution of evaporation performance is analyzed. [Fig. 8](#) shows the evolution of the interfacial latent and sensible heat fluxes along the evaporation surface. The latent heat flux decreases and the absolute

Table 1
Accuracy, range and resolution of the measurement instruments.

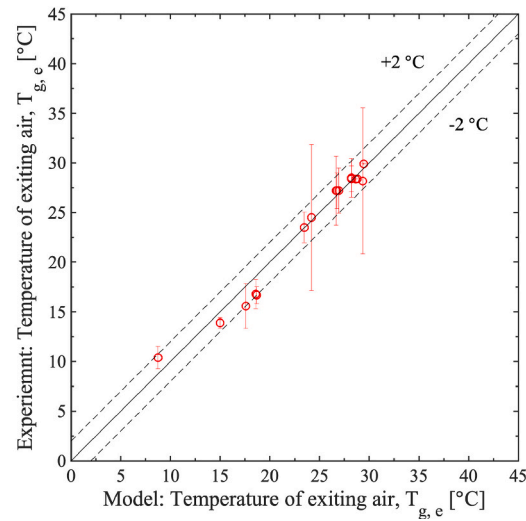
Instrument	Accuracy	Range	Resolution
Telaire T9602 K-Type Thermocouple	±0.5 °C	−20 to +70 °C	0.01 °C
Telaire T9602 Relative humidity Sensor	±2%	20 to 80%	0.01%
UNI-T UT363 Anemometer	±5%	0 to 30 m/s	0.1 m/s



(a) Evaporation rate per surface.



(b) Exiting air relative humidity.



(c) Exiting air temperature.

Fig. 6. Comparison of model-predicted and experimental results using the CEE setup of Fig. 5. Dashed lines indicate bounds on the deviation between the model and experimental result. Error bars represent the standard deviation in the data set at each operating condition. Operating conditions and the number of repetitions can be found in Appendix B.

Table 2
Summary of simulation base case values.

Variable	Value
Ambient temperature ($T_{g, in}$)	10 °C
Ambient relative humidity ($R_{h, in}$)	60%
Liquid inlet temperature ($T_{l, in}$)	10 °C
Liquid flow rate ($\dot{m}_{l, in}$)	6 L/hr/m ² _{surface}
Air flow speed (u)	4 m/s
Surface dimensions ($L \times W$)	1 × 1 m ²
Spacing between surfaces (d)	2 cm
Liquid inlet salinity (S_{in})	0 g/kg

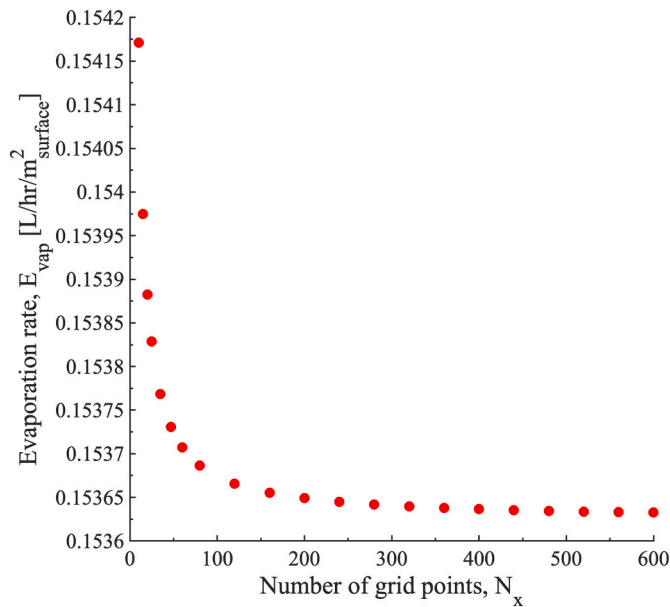


Fig. 7. Results of the grid independence study. Grid independent results were obtained at 500 nodes or higher.

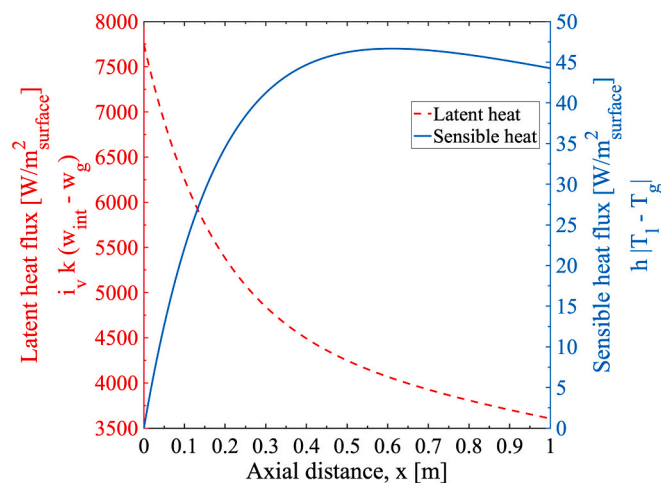


Fig. 8. Evolution of latent and sensible heat fluxes along the evaporation surface at base case conditions. Latent heat flux dominates sensible heat flux at all locations.

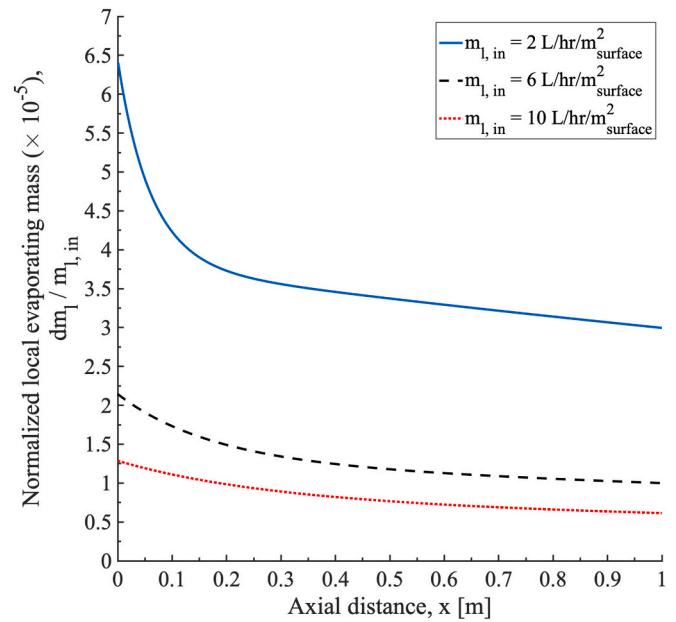


Fig. 9. Local evaporating mass along the evaporation surface at different liquid inlet flow rates.

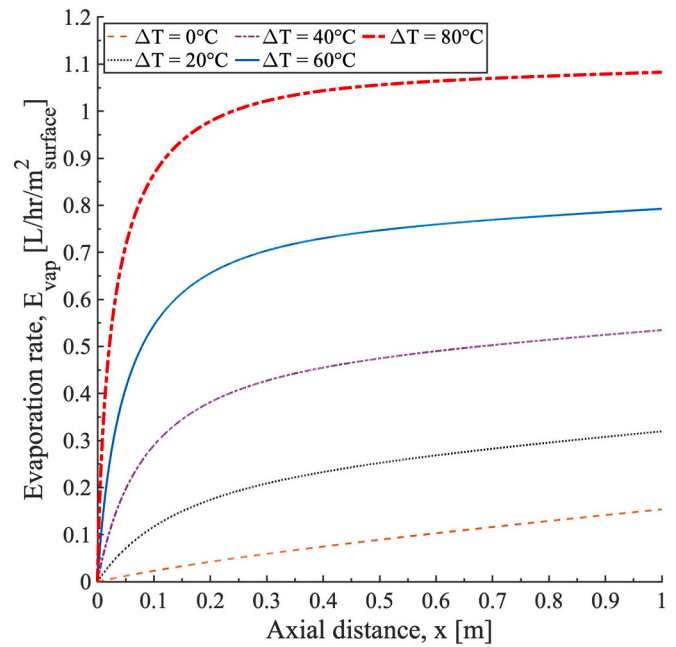


Fig. 10. Accumulated evaporation rate (per unit surface area) along the evaporation surface length at different values of temperature difference.

value of the sensible heat flux increases along the flow path. This is to be expected; as the air becomes saturated, the latent heat flux is decreased. Meanwhile, the liquid film releases latent heat, increasing the temperature difference between the liquid and air, increasing the sensible heat flux. It is observed that the latent heat dominates the sensible heat; it is approximately 90 times greater than the sensible heat in the middle of the surface and approximately 80 times greater at the end. It can be concluded that the key driving force for evaporation is not the difference in temperature between the liquid surface and the air, but rather the difference in the vapor pressure between the liquid surface and the air, which confirms the experimental observation found by Lu et al. [29]. Heating the liquid increases the vapor pressure gradient and thus

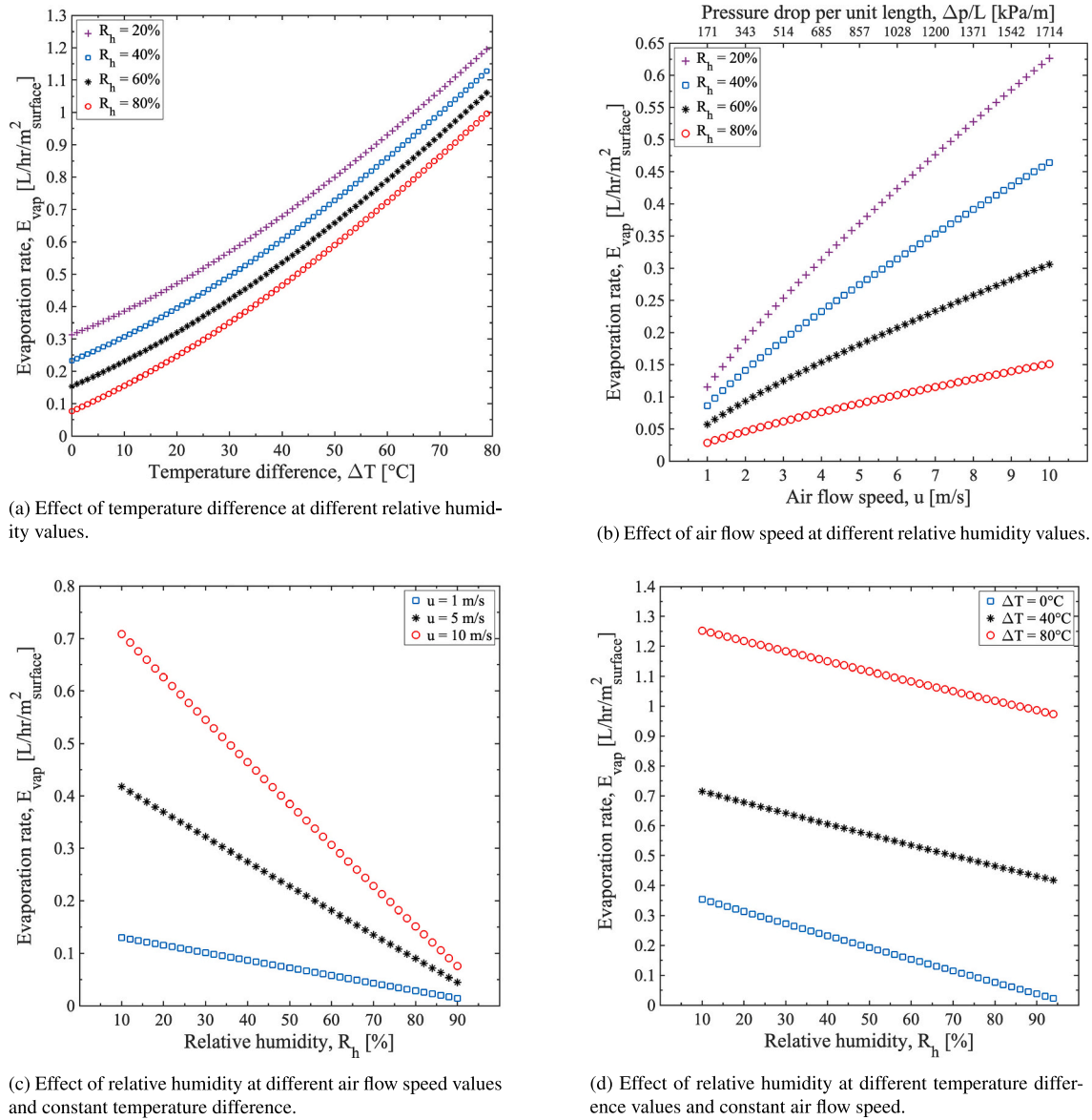


Fig. 11. Effect of temperature difference, air flow speed, and ambient relative humidity on total evaporation rate per surface.

increases the moisture content gradient (Eq. (12)), which in turns increases the evaporation rate. In contrast, heating the air would decrease this gradient, resulting in lower evaporation rate. Therefore, heating the liquid is more beneficial than heating the air to increase evaporation rate.

Fig. 9 shows the local evaporation mass flow rate, normalized by the inlet liquid mass flow rate at three different inlet flow rate values. It is observed that the normalized evaporating mass decreases along the evaporation surface. It is also observed that the highest normalized evaporation occurs with lower liquid inlet flow rate; however, the highest absolute evaporation occurs with higher liquid inlet flow rate. This is justified by the decrease in latent heat flux downstream (Fig. 8).

Fig. 10 shows the spatial profile of the accumulated evaporation rate along the evaporation surface at five different values of temperature difference. The accumulated evaporation rate increases in the flow direction as water evaporates from the film. The accumulated evaporation rate has a linear profile at $\Delta T = 0^\circ\text{C}$, and becomes logarithmic as ΔT increases. At high temperature difference ($\Delta T = 80^\circ\text{C}$), evaporation rate is high at the beginning of the surface, then approaches a horizontal asymptote of approximately $1.1 \text{ L/h/m}^2_{\text{surface}}$ after an axial distance of 0.5 m . The liquid film cools down as it flow over the surface (as a portion

of the film evaporates), resulting in lower vapor pressure and lower latent heat flux. It is concluded that shorter evaporation surfaces should be utilized at high liquid temperatures to maintain high evaporation performance and best utilize the available surface area. The effect of preheating the liquid is discussed in more detail in the next section.

4.2. Influence of liquid preheating

Simulations were carried out to investigate the effect of liquid preheating on the total evaporation rate per surface. Liquid inlet temperature was varied from 10°C to 90°C , at three different ambient relative humidity values (40%, 60%, 80%) while maintaining other variables at the base case (Table 2). Results are shown in Fig. 11a. Evaporation rate increases in a near linear trend as the temperature difference increases. Increasing the temperature difference from 0°C to 80°C at relative humidity of 80% increases the evaporation rate 13 times from $0.08 \text{ L/h/m}^2_{\text{surface}}$ to $1.00 \text{ L/h/m}^2_{\text{surface}}$, while at relative humidity of 40%, the evaporation rate increases 5 times from $0.23 \text{ L/h/m}^2_{\text{surface}}$ to $1.13 \text{ L/h/m}^2_{\text{surface}}$. This is due to the increase in the saturation vapor pressure and moisture content at the interface as liquid temperature increases, which in turn increases the driving force for mass transfer, as shown in Eq. (12),

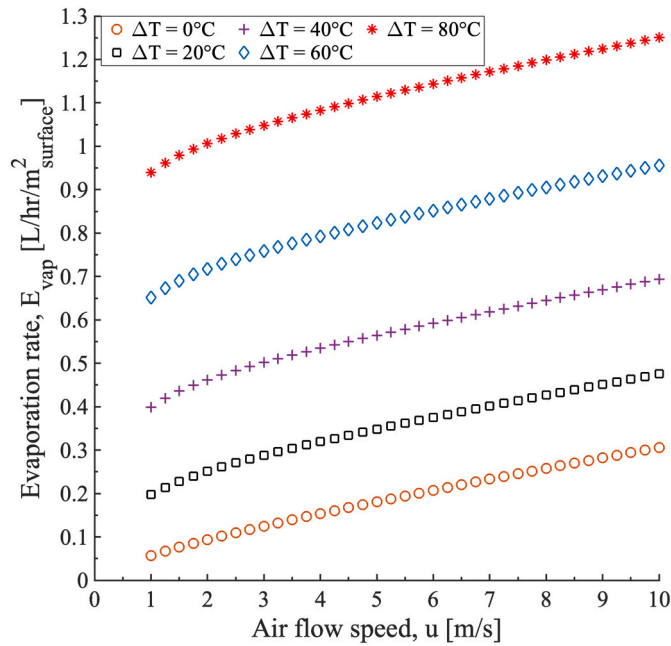


Fig. 12. Total evaporation rate per surface as a function of air flow speed at different values of temperature difference.

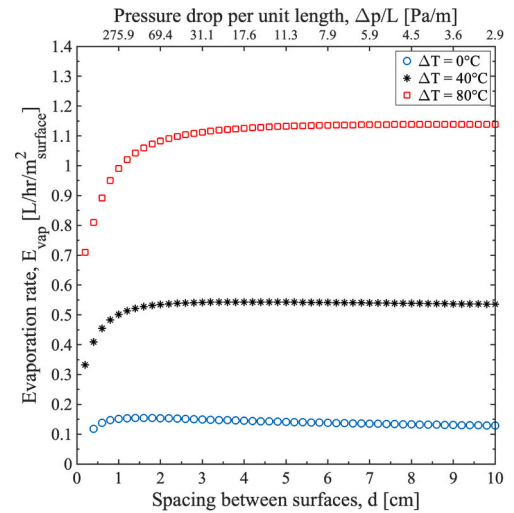
and as observed by Zamen et al. [19]. On the other hand, as the relative humidity increases from 40% to 80%, the moisture content in the flowing air increases, thus decreasing the driving force for mass transfer (Eq. (12)). This justifies the decrease in evaporation rates at higher ambient relative humidity.

4.3. Influence of air flow speed

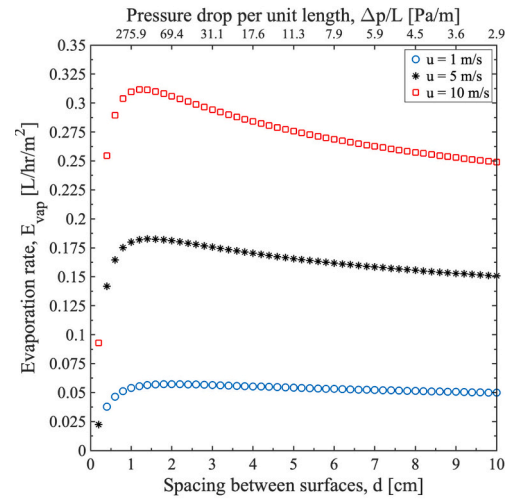
Simulations were carried out to investigate the effect of air flow speed on the total evaporation rate per surface. Air flow speed was varied from 1 m/s to 10 m/s, at three different ambient relative humidity values (40%, 60%, 80%) while maintaining other variables at the base case (Table 2). Results are shown in Fig. 11b. As expected, as the relative humidity increases, the evaporation rate decreases (Eq. (12)). Increasing air flow speed increases the evaporation rate. At relative humidity of 40%, as the air flow speed increases from 1 m/s to 10 m/s, the evaporation rate increases 5 times from 0.086 L/h/m² surface to 0.46 L/h/m² surface. Similar increase is observed at relative humidity of 80% (0.028 L/h/m² surface to 0.15 L/h/m² surface).

While higher air flow speed enhances evaporation, high speeds also act to increase the pressure drop along the surface (Eq. (31)), resulting in higher fan power consumption. Pressure drop increases in a linear fashion from 171 kPa/m at air flow speed of 1 m/s to 1714 kPa/m at speed of 10 m/s. The trade-off between the increase in evaporation rate per surface (decreasing capital cost) and the increase in pressure drop (increasing operational cost) must be considered.

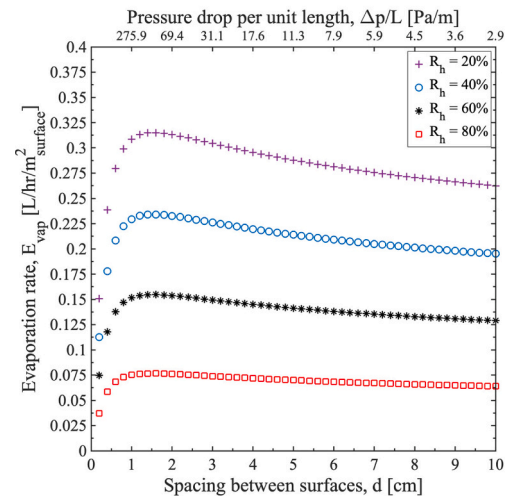
To visualize the relative benefits of liquid preheating and air flow speed, the total evaporation rate per surface was plotted as a function of air flow speed at different liquid inlet temperatures in Fig. 12. It is clear that increasing the temperature difference between inlet liquid and air has greater enhancement on evaporation rate than increasing the air flow speed. As the temperature difference is doubled from 40 °C to 80 °C, the evaporation rate is doubled from 0.53 L/h/m² surface to 1.08 L/h/m² surface at a speed of 4 m/s, and increases by factor of 1.8 from 0.64 L/h/m² surface to 1.20 L/h/m² surface at a speed of 8 m/s. On the other hand, as the air flow speed is doubled from 4 m/s to 8 m/s, the evaporation rate increases by just 20% from 0.53 L/h/m² surface to 0.64 L/h/m² surface at $\Delta T=40^\circ\text{C}$, and increases by just 11% at $\Delta T=80^\circ\text{C}$. Nevertheless, increased air flow speed should still be considered as a method of



(a) Varying temperature difference values.



(b) Varying air flow speeds.



(c) Varying ambient relative humidity values.

Fig. 13. Total evaporation rate per surface as a function of the spacing between evaporation surfaces.

increasing evaporation performance. Selecting an optimal air flow speed and temperature difference will require information about the cost of thermal versus electrical energy at the installation site of interest.

4.4. Influence of relative humidity

Simulations were carried out to investigate the effect of relative humidity on the total evaporation rate per surface. Relative humidity was varied from 10% to 90%, at three different air flow speeds (1 m/s, 5 m/s, 10 m/s) and three different temperature difference values (0 °C, 40 °C, 80 °C) while maintaining other variables at the base case (Table 2). It is observed from Fig. 11c that at all air flow speeds, as the relative humidity increases from 10% to 90%, the evaporation rates drop sharply by a factor of 10. For example, at air flow speed of 10 m/s, the evaporation rate drops from 0.7 L/h/m²_{surface} to 0.075 L/h/m²_{surface}, while at air flow speed of 1 m/s, it drops from 0.13 L/h/m²_{surface} to 0.014 L/h/m²_{surface}. In contrast, we see a linear relationship between relative humidity and evaporation rate, with nearly equal slopes for all values of temperature difference (Fig. 11d). As a result, at $\Delta T=80$ °C, as the relative humidity increases from 10% to 90%, the evaporation rate drops by just 22% from 1.25 L/h/m²_{surface} to 0.97 L/h/m²_{surface}, while at $\Delta T=0$ °C, it drops sharply by 93% from 0.35 L/h/m²_{surface} to 0.022 L/h/m²_{surface}. It can be concluded that increasing the temperature difference by preheating the liquid is a good strategy to overcome the negative effect of high humidity.

4.5. Influence of spacing between evaporation surfaces

Simulations were carried out to investigate the effect of spacing between evaporation surfaces on the total evaporation rate per surface. Spacing was varied from 1 cm to 10 cm, at three different temperature differences (0 °C, 40 °C, 80 °C), three different ambient relative humidity values (40%, 60%, 80%) and three different air flow speeds (1 m/s, 5 m/s, 10 m/s) while maintaining other variables at the base case (Table 2). Results are shown in Fig. 13.

Fig. 13a shows the effect of spacing at different values of temperature difference. At $\Delta T=0$ °C and $\Delta T=40$ °C, the peak in evaporation rate is observed at spacing greater than 1.2 cm; for $\Delta T=0$ °C, evaporation rate slightly drops as spacing increases above 2 cm. For $\Delta T=80$ °C, the

maximum evaporation rate is observed at spacing greater than 2.4 cm.

Fig. 13b shows the effect of spacing at different air flow speeds. It is observed that at low speed (1 m/s), the maximum evaporation rate is obtained at spacing greater than 1 cm. At higher speeds (5 m/s and 10 m/s), the peak is achieved at spacing values ranging from 0.8 cm to 2.6 cm, then evaporation rate drops as spacing increases. Fig. 13c shows the effect of spacing at different ambient relative humidity values. At ambient relative humidity of 80%, the peak is observed at spacing greater than 1 cm; for humidity values of 60% and 40%, the peak is observed between 1 cm and 2 cm, then evaporation rate drops as spacing increases.

It is concluded that for most operating conditions, maximum

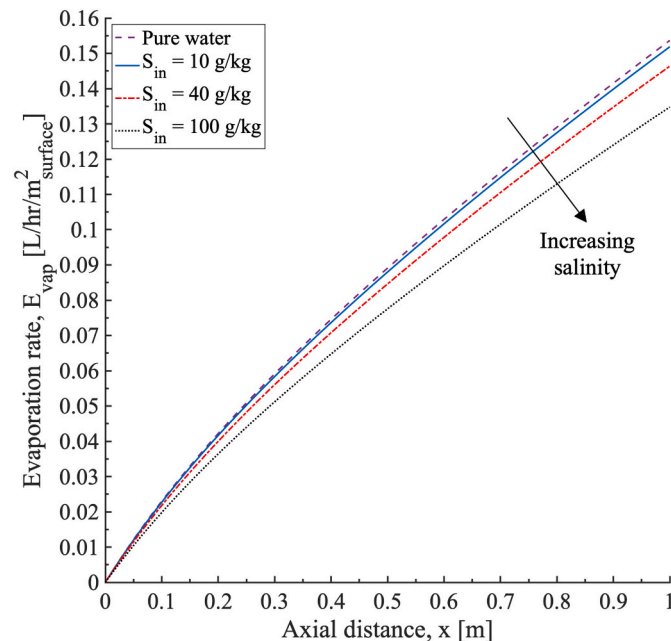
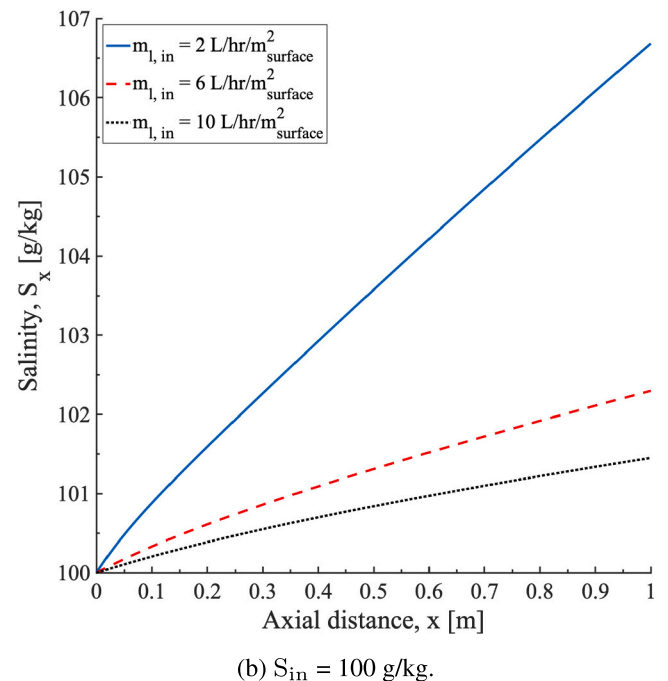
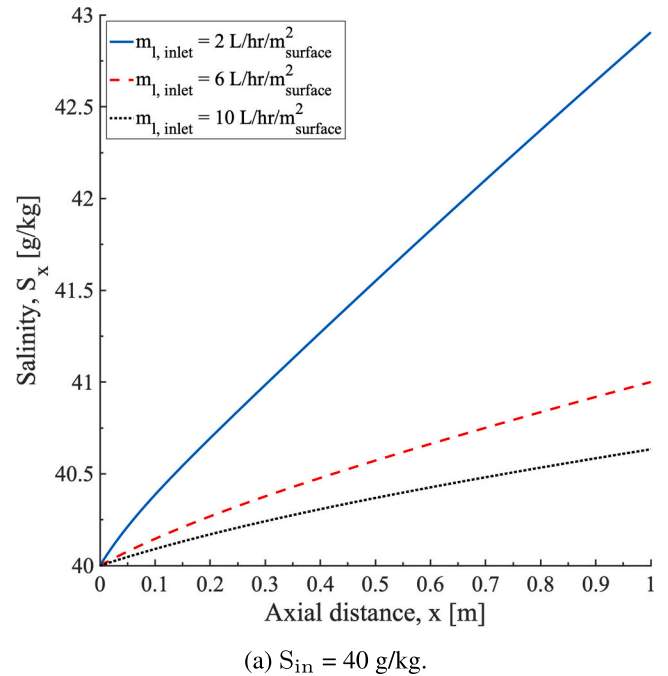
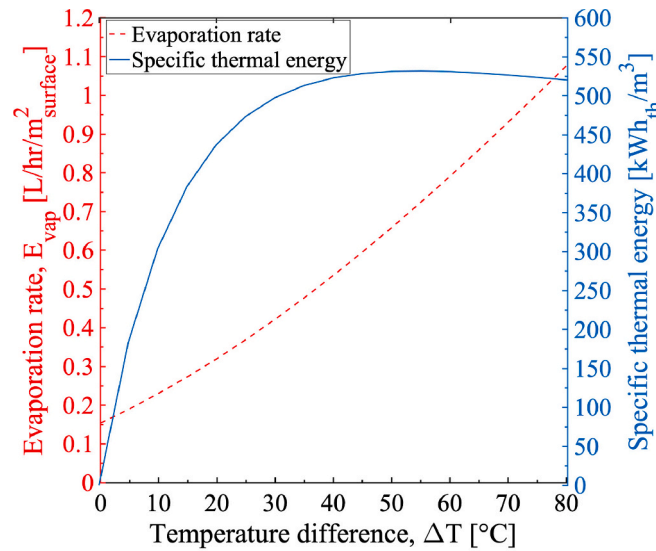
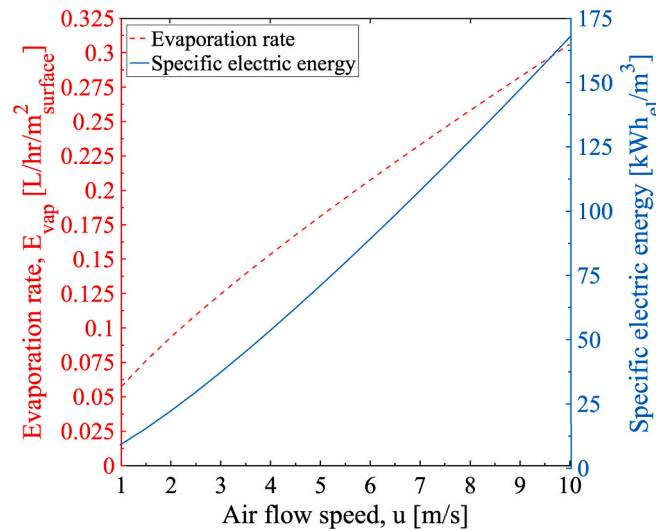


Fig. 14. Accumulated evaporation rate along the evaporation surface at different inlet salinity values.

Fig. 15. Salinity variation along the evaporation surface at different values of liquid inlet flow rate.



(a) Specific thermal energy consumption.



(b) Specific electric energy consumption.

Fig. 16. Trade-off between evaporation rate and specific thermal (a) and electric (b) energy consumption associated with liquid preheating and fan operation, respectively.

evaporation is achieved with surface spacing between 1 cm and 2 cm; the exception is for high temperature difference ($\Delta T=80$ °C) where maximum evaporation occurs when spacing is greater than 2.4 cm. Surface spacing will also affect system pressure drop, with pressure drop decreasing as the spacing increases (Eq. (31)). Pressure drop decreases from 276 Pa/m at spacing of 1 cm to 69.5 Pa/m at spacing of 2 cm, which, depending on the specific fan chosen, could result in a 4 times decrease in required fan power. Finally, we note that surface spacing has an overall lower effect on the evaporation rate than preheating the liquid and air flow speed (Fig. 11a and b).

4.6. Influence of liquid salinity

Simulations were carried out to investigate the effect of liquid inlet salinity on the total evaporation rate per surface. Salinity values (S_{in}) of 0, 10, 40, and 100 g/kg are of interest because they represent pure water, brackish water, seawater, and highly saline liquid brine,

respectively. Fig. 14 shows the simulation results for these salinity values while maintaining other variables at the base case (Table 2). As salinity increases, the evaporation rate decreases. At low salinity (brackish water, 10 g/kg), saline liquid behaves similar to pure water, while as the salinity is increased to 40 g/kg (seawater) and 100 g/kg (highly saline liquid brine), the evaporation rate decreases by 4.8% and by 12.3%, respectively. This is due to the decrease in the latent heat of vaporization as salinity increases [26], which in turn decreases the latent heat flux. Consequently, the evaporation rate decreases due to the fact that the evaporation process is dominated by latent heat flux (Fig. 8). Fig. 15 shows the variation of local salinity (S_x) along the evaporation surface at three different liquid inlet flow rates (2, 6, 10 L/h/m² surface). As expected, local salinity increases along the surface as water evaporates. The rate of increase in local salinity is higher with lower liquid inlet flow rate. This is because the ratio of evaporated mass flux to inlet mass flow rate increases as inlet flow rate increases (Fig. 9). This in turn causes the concentration (or local salinity) of the liquid to

increase. For inlet salinity of 40 g/kg, the local salinity increases by 7% at a flow rate of 2 L/h/m²_{surface} and by 1.6% at a flow rate of 10 L/h/m²_{surface}. Similarly, for inlet salinity of 100 g/kg, salinity increases by 6.7% at a flow rate of 2 L/h/m²_{surface} and by 1.4% at a flow rate of 10 L/h/m²_{surface}. It is observed that, at a given flow rate, the percent increase in salinity (S_x) is similar at all inlet salinity values (S_{in}).

4.7. Energy consumption

Simulations were carried out to investigate the effect of liquid preheating and air flow speed on the energy consumption of the system. Liquid inlet temperature was varied from 10 °C to 90 °C and air flow speed was varied from 1 m/s to 10 m/s while maintaining other variables at the base case (Table 2). Results are shown in Fig. 16a and b, respectively.

Increasing the temperature difference increases the thermal energy consumption per evaporated volume (Fig. 16a). Increasing the temperature difference from 10 °C to 40 °C increases the evaporation rate by a factor of 2.3 (from 0.23 L/h/m²_{surface} to 0.54 L/h/m²_{surface}), while the specific thermal energy consumption increases by a factor of 1.74 (from 303 kWh_{th}/m³ to 523 kWh_{th}/m³). Increasing the temperature difference from 40 °C to 80 °C doubles the evaporation rate (from 0.54 L/h/m²_{surface} to 1.08 L/h/m²_{surface}), while the specific thermal energy consumption stays approximately at constant value (520 kWh_{th}/m³). Operating at a temperature difference ranging from 40 °C to 80 °C is recommended because the specific energy consumption remains constant in this range, while the evaporation rate increases significantly resulting in a decrease in the required surface area and an associated decrease in the capital cost. The thermal energy required for liquid preheating could be supplied from several sources including process waste heat, concentrated solar collectors, etc. A cost-optimal system would consider the trade-off between increased evaporation rate (resulting in a decrease in capital cost) and increase in required thermal energy (increase in operational cost).

Increasing the air flow speed increases the electric energy consumption per evaporated volume (Fig. 16b). At air flow speed of 1 m/s, the evaporation rate is 0.06 L/h/m²_{surface} and the specific electric energy consumption is 9 kWh_{el}/m³. As the air flow speed increases to 5 m/s, the evaporation rate increases by a factor of 3 (from 0.06 L/h/m²_{surface} to 0.18 L/h/m²_{surface}) and the specific electric energy consumption increases to 71 kWh_{el}/m³. Increasing the air flow speed from 5 m/s to 10 m/s increases the evaporation rate by a factor of 1.7 (from 0.18 L/h/m²_{surface} to 0.31 L/h/m²_{surface}), while the specific electric energy consumption increases by a factor of 2.4 (from 71 kWh_{el}/m³ to 168 kWh_{el}/m³). The specific electric energy consumption is low relative to the specific thermal energy consumption. The cost of locally available energy sources will need to be considered in order to design a cost-optimized system.

5. Conclusions

This work proposes a novel convection-enhanced evaporation (CEE) system consisting of horizontally packed evaporation surfaces. The presented mathematical model combines the effect of several operating conditions including ambient conditions (temperature and humidity), liquid inlet conditions (temperature, flow rate and salinity), and operating conditions (air flow speed and surface spacing). Comparison between model-predicted evaporation rate and experimental results show good agreement; for 11 out of 16 operating conditions relative error was within 10%, for 4 conditions relative error was within 20%, and for 1 condition, relative error was within 35%. Relative humidity and temperature of the exit air agreed with the experiments to within 3% and 2 °C, respectively. The effect of several operating conditions on the evaporation performance is investigated. The following conclusions are drawn:

1. Liquid inlet temperature and air flow speed have a larger effect on evaporation rate than ambient temperature and humidity, spacing between surfaces, and liquid inlet salinity.
2. Increasing the inlet temperature difference between the liquid and the air increases the evaporation rate by up to a factor of 13.
3. Increasing the air flow speed increases the evaporation rate by up to a factor of 5. However, an increase in air flow speed also results in an increase in pressure drop (electric energy consumption). The trade-off between enhanced evaporation rate and energy demand must be analyzed based on installation conditions.
4. In humid weather, it is recommended to preheat the liquid above ambient temperature to overcome the negative effect of high relative humidity.
5. Maximum evaporation is obtained when surface spacing is between 1 cm and 2 cm for most operating conditions.
6. Evaporation rate decreases as liquid salinity increases. The evaporation rates for seawater (40 g/kg) and highly saline brine (100 g/kg) are 4.8% and 12.3% less than that of pure water, respectively.

The presented mathematical model and resulting design guidelines provide the basis for further investigation of convection-enhanced evaporation (CEE) systems. Optimization of the operating conditions and experimental testing of the proposed CEE system with real brines are parts of future work.

Nomenclature

A	evaporation surface area, m ²
c_p	specific heat, J kg ⁻¹ K ⁻¹
d	vertical spacing between surfaces, m
D_h	hydraulic diameter, m
e	specific internal energy, J kg ⁻¹
E_{vap}	evaporation rate, kg s ⁻¹
$\mathbf{f}$	body force vector, m s ⁻²
f	Darcy friction factor
h	heat transfer coefficient, W m ⁻² K ⁻¹
i	specific enthalpy, J kg ⁻¹
i_{fgw0}	latent heat of water at 0 °C, J kg ⁻¹
k	mass transfer coefficient, kg m ⁻² s ⁻¹
L	evaporation surface length, m
Le_f	Lewis factor
$\dot{m}$	mass flow rate, kg s ⁻¹
N_s	number of evaporation surfaces
N_x	number of grid points
Pr	Prandtl number
p	pressure, Pa
P_{el}	electric power consumption, W
P_{th}	thermal power consumption, W
p_{atm}	atmospheric pressure, Pa
p_{sat}	saturation vapor pressure, Pa
Δp	pressure drop, Pa
$\mathbf{q}$	heat transfer vector, J s ⁻¹
$\mathbf{q}''$	heat transfer flux vector, J m ⁻² s ⁻¹
$\dot{q}$	volumetric heat generation, J m ⁻³ s ⁻¹
Re	Reynolds number
S	Salinity, g/kg
t	time, s
$\mathbf{T}$	stress tensor, kg m ⁻¹ s ⁻²
T	temperature, K
Δt	experiment duration, s
ΔT	temperature difference
$\mathbf{u}$	velocity vector, m/ s
u	axial velocity component, m/ s
$\dot{V}$	volumetric flow rate, m ³ s ⁻¹
ΔV	volume change, m

W evaporation surface width, m
 x axial distance, m

Greek symbols

α dimensionless shape factor
 η_{fan} fan efficiency, %
 λ thermal conductivity, $W\ m^{-1}\ K^{-1}$
 ν kinematic viscosity, $m^2\ s^{-1}$
 ρ density, $kg\ m^{-3}$
 τ shear stress tensor, $kg\ m^{-1}\ s^{-2}$
 ω moisture content, $kg\ kg^{-1}$

Subscripts

a dry air
 e exit conditions
 g gas phase (air-vapor mixture)
 in inlet conditions
 int liquid-air interface
 l liquid phase
 v water vapor
 x conditions at location x

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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The authors declare the following interest: P.B. and his team at QuadSun Solar, Ltd. manufacture the CEE systems used for experimental testing in this study. This interest has not significantly influenced the presentation or interpretation of the data; model development occurred prior to receiving experimental test results for comparison.

Author statement

Mustafa F. Kaddoura: Conceptualization, Methodology, Software, Validation, Formal analysis, Writing-Original Draft, Visualization.

Matthew Chosa: Validation, Writing – Review & Editing.

Prakash Bhalekar: Validation, Investigation.

Natasha C. Wright: Conceptualization, Methodology, Writing – Review & Editing, Supervision.

Appendix A. Runge-Kutta method implementation

The fourth order Runge-Kutta method utilizes weighted average of several solutions over the interval Δx in order to improve accuracy [30].

Eqs. (12), (13), (14) and (18) can be expressed as,

$$\frac{d\dot{m}_l}{dx} = F(\dot{m}_l, \omega_g, i_g, T_l) \quad (A.1)$$

$$\frac{d\omega_g}{dx} = G(\dot{m}_l, \omega_g, i_g, T_l) \quad (A.2)$$

$$\frac{di_g}{dx} = H(\dot{m}_l, \omega_g, i_g, T_l) \quad (A.3)$$

$$\frac{dT_l}{dx} = I(\dot{m}_l, \omega_g, i_g, T_l) \quad (A.4)$$

The values of the four variables $(\dot{m}_l, \omega_g, i_g, T_l)$ at position $(i + 1)$ can be determined using the following,

$$\dot{m}_l^{i+1} = \dot{m}_l^i + (\dot{j}_1^{i+1} + 2\dot{j}_2^{i+1} + 2\dot{j}_3^{i+1} + \dot{j}_4^{i+1}) / 6 \quad (A.5)$$

$$\omega_g^{i+1} = \omega_g^i + (k_1^{i+1} + 2k_2^{i+1} + 2k_3^{i+1} + k_4^{i+1}) / 6 \quad (A.6)$$

$$i_g^{i+1} = i_g^i + (\dot{l}_1^{i+1} + 2\dot{l}_2^{i+1} + 2\dot{l}_3^{i+1} + \dot{l}_4^{i+1}) / 6 \quad (A.7)$$

$$T_l^{i+1} = T_l^i + (o_1^{i+1} + 2o_2^{i+1} + 2o_3^{i+1} + o_4^{i+1}) / 6 \quad (A.8)$$

where

$$\dot{j}_1^{i+1} = \Delta x \cdot F(\dot{m}_l^i, \omega_g^i, i_g^i, T_l^i) \quad (A.9)$$

$$k_1^{i+1} = \Delta x \cdot G(\dot{m}_l^i, \omega_g^i, i_g^i, T_l^i) \quad (A.10)$$

$$\dot{l}_1^{i+1} = \Delta x \cdot H(\dot{m}_l^i, \omega_g^i, i_g^i, T_l^i) \quad (A.11)$$

$$o_1^{i+1} = \Delta x \cdot I \left(\dot{m}_l^i, \omega_g^i, i_g^i, T_l^i \right) \quad (\text{A.12})$$

$$j_2^{i+1} = \Delta x \cdot F \left(\dot{m}_l^i + \frac{j_1^{i+1}}{2}, \omega_g^i + \frac{k_1^{i+1}}{2}, i_g^i + \frac{l_1^{i+1}}{2}, T_l^i + \frac{o_1^{i+1}}{2} \right) \quad (\text{A.13})$$

$$k_2^{i+1} = \Delta x \cdot G \left(\dot{m}_l^i + \frac{j_1^{i+1}}{2}, \omega_g^i + \frac{k_1^{i+1}}{2}, i_g^i + \frac{l_1^{i+1}}{2}, T_l^i + \frac{o_1^{i+1}}{2} \right) \quad (\text{A.14})$$

$$l_2^{i+1} = \Delta x \cdot H \left(\dot{m}_l^i + \frac{j_1^{i+1}}{2}, \omega_g^i + \frac{k_1^{i+1}}{2}, i_g^i + \frac{l_1^{i+1}}{2}, T_l^i + \frac{o_1^{i+1}}{2} \right) \quad (\text{A.15})$$

$$o_2^{i+1} = \Delta x \cdot I \left(\dot{m}_l^i + \frac{j_1^{i+1}}{2}, \omega_g^i + \frac{k_1^{i+1}}{2}, i_g^i + \frac{l_1^{i+1}}{2}, T_l^i + \frac{o_1^{i+1}}{2} \right) \quad (\text{A.16})$$

$$j_3^{i+1} = \Delta x \cdot F \left(\dot{m}_l^i + \frac{j_3^{i+1}}{2}, \omega_g^i + \frac{k_3^{i+1}}{2}, i_g^i + \frac{l_3^{i+1}}{2}, T_l^i + \frac{o_3^{i+1}}{2} \right) \quad (\text{A.17})$$

$$k_3^{i+1} = \Delta x \cdot G \left(\dot{m}_l^i + \frac{j_3^{i+1}}{2}, \omega_g^i + \frac{k_3^{i+1}}{2}, i_g^i + \frac{l_3^{i+1}}{2}, T_l^i + \frac{o_3^{i+1}}{2} \right) \quad (\text{A.18})$$

$$l_3^{i+1} = \Delta x \cdot H \left(\dot{m}_l^i + \frac{j_3^{i+1}}{2}, \omega_g^i + \frac{k_3^{i+1}}{2}, i_g^i + \frac{l_3^{i+1}}{2}, T_l^i + \frac{o_3^{i+1}}{2} \right) \quad (\text{A.19})$$

$$o_3^{i+1} = \Delta x \cdot I \left(\dot{m}_l^i + \frac{j_3^{i+1}}{2}, \omega_g^i + \frac{k_3^{i+1}}{2}, i_g^i + \frac{l_3^{i+1}}{2}, T_l^i + \frac{o_3^{i+1}}{2} \right) \quad (\text{A.20})$$

$$j_4^{i+1} = \Delta x \cdot F \left(\dot{m}_l^i + \frac{j_3^{i+1}}{2}, \omega_g^i + \frac{k_3^{i+1}}{2}, i_g^i + \frac{l_3^{i+1}}{2}, T_l^i + \frac{o_3^{i+1}}{2} \right) \quad (\text{A.21})$$

$$k_4^{i+1} = \Delta x \cdot G \left(\dot{m}_l^i + \frac{j_3^{i+1}}{2}, \omega_g^i + \frac{k_3^{i+1}}{2}, i_g^i + \frac{l_3^{i+1}}{2}, T_l^i + \frac{o_3^{i+1}}{2} \right) \quad (\text{A.22})$$

$$l_4^{i+1} = \Delta x \cdot H \left(\dot{m}_l^i + \frac{j_3^{i+1}}{2}, \omega_g^i + \frac{k_3^{i+1}}{2}, i_g^i + \frac{l_3^{i+1}}{2}, T_l^i + \frac{o_3^{i+1}}{2} \right) \quad (\text{A.23})$$

$$o_4^{i+1} = \Delta x \cdot I \left(\dot{m}_l^i + \frac{j_3^{i+1}}{2}, \omega_g^i + \frac{k_3^{i+1}}{2}, i_g^i + \frac{l_3^{i+1}}{2}, T_l^i + \frac{o_3^{i+1}}{2} \right) \quad (\text{A.24})$$

Appendix B. Experimental conditions and results

Summary of the experiment conditions and comparison between the experiment and model results are presented in [Tables B.1 and B.2](#) respectively.

Table B.1

Details of experiment conditions.

Exp. no.	Liquid flow	Liquid inlet	Ambient relative	Ambient	Air flow	Surface grade	Repetitions
	rate,	temperature,	humidity,	temperature,	speed,		
	$\dot{m}_{l,in}$	$T_{l,in}$	$R_{h,in}$	$T_{g,in}$	u		
	(L/h/m ² _{surface})	(°C)	(%)	(°C)	(m/s)		
1	2.10	25.17	57.40	27.05	10.0	Nylon PA-6 10% mineral filled	3
2	6.50	13.85	72.73	15.18	8.7	Nylon PA-6 10% mineral filled	3
3	2.60	60.00	80.00	8.20	6.0	Nylon PA-6 10% mineral filled	3
4	3.35	38.00	48.97	23.70	6.0	Nylon PA-6 10% mineral filled	4
5	6.50	13.63	55.78	17.95	10.0	Nylon PA-6 10% mineral filled	5
6	6.50	16.06	45.50	17.89	8.7	Nylon PA-6 10% mineral filled	5
7	6.50	13.46	40.43	18.98	10.0	Nylon PA-6 10% mineral filled	7
8	1.50	66.37	68.25	29.95	10.0	Nylon PA-6 10% mineral filled	3
9	1.87	27.20	75.32	28.45	10.0	Polypropylene	8
10	0.86	61.83	71.86	28.83	6.0	Polypropylene	3
11	1.23	61.21	69.66	29.00	6.0	Polypropylene	3
12	1.30	38.00	54.48	28.63	10.0	Nylon PA-6	3
13	2.03	38.00	53.15	27.33	10.0	Nylon PA-6	4
14	3.00	27.16	41.81	29.99	8.7	Nylon PA-6	3
15	2.03	38.00	53.15	27.33	10.0	Nylon PA-6	4
16	2.97	26.50	51.82	27.22	9.0	Nylon PA-6	4

Table B.2

Comparison between experiment and model results.

Exp. no.	Experiment	Model	Experiment exiting	Model exiting	Experiment exiting	Model exiting
	evaporation rate,	evaporation rate,	relative humidity,	relative humidity,	air temperature,	air temperature,
	E_{vap}	E_{vap}	$R_{h, e}$	$R_{h, e}$	$T_{g, e}$	$T_{g, e}$
	($\text{L/h/m}^2_{\text{surface}}$)	($\text{L/h/m}^2_{\text{surface}}$) (% error)	(%)	(%)	(°C)	(°C)
1	0.236	0.2202 (6.7%)	57.67	60.44	27.20	26.63
2	0.083	0.0848 (−2.16%)	76.03	75.02	13.90	15.01
3	0.258	0.2696 (−4.5%)	84.00	87.54	10.40	8.74
4	0.277	0.2733 (1.33%)	49.21	53.77	23.50	23.46
5	0.188	0.1559 (17.1%)	61.82	59.04	15.61	17.60
6	0.203	0.1966 (3.15%)	52.66	47.88	15.69	18.59
7	0.223	0.2257 (−1.21%)	46.52	45.07	16.67	18.67
8	0.303	0.2676 (11.68%)	69.00	73.24	29.90	29.43
9	0.117	0.1275 (−8.97%)	75.79	77.23	28.35	28.21
10	0.124	0.1649 (−32.98%)	72.13	74.56	28.36	28.61
11	0.194	0.2018 (−4.02%)	70.41	72.66	28.40	28.79
12	0.281	0.2731 (2.81%)	54.71	57.73	28.46	28.19
13	0.340	0.2978 (12.41%)	53.50	56.58	27.16	26.92
14	0.308	0.3123 (−1.39%)	46.60	45.67	28.18	29.34
15	0.340	0.2978 (12.41%)	53.50	56.58	27.16	26.92
16	0.241	0.2475 (−2.69%)	52.18	55.27	27.17	26.75

Appendix C. Properties of saline water

The specific heat of saline water is computed as [31]

$$c_p = (A + BT + CT^2 + DT^3) \times 10^3, \quad (\text{C.1})$$

where

$$\begin{aligned} A &= 5.328 - 9.76 \times 10^{-2} S + 4.04 \times 10^{-4} S^2 \\ B &= -6.913 \times 10^{-3} + 7.351 \times 10^{-4} S - 3.15 \times 10^{-6} S^2 \\ C &= 9.6 \times 10^{-6} - 1.927 \times 10^{-6} S + 8.23 \times 10^{-9} S^2 \\ D &= 2.5 \times 10^{-9} + 1.666 \times 10^{-9} S - 7.125 \times 10^{-12} S^2 \end{aligned}$$

The latent heat of vaporization of saline water is computed as [26]

$$i_v = i_{v, \text{pure water}} \times \left(1 - \frac{S}{1000}\right), \quad (\text{C.2})$$

where i_v and $i_{v, \text{pure water}}$ are the latent heat of vaporization of saline water and pure water, respectively.

The vapor pressure of saline water is computed as [32]

$$\log_{10} \left(\frac{P_v}{P_{v, \text{pure water}}} \right) = -2.1609 \times 10^{-4} S - 3.5012 \times 10^{-7} S^2, \quad (\text{C.3})$$

where P_v and $P_{v, \text{pure water}}$ are the vapor pressures of saline water and pure water, respectively.

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